FLEET TEAM DIGEST

737MAX-FTD-21-17001

Issue Title : Cabin Pressurization Digital Selector Panel (DSP) Blanking/Dimming Issue

Airplane Model	ATA	Minor Model(s)
737MAX	2131-02	-7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
	9E991	

Originated Date	Last Revised Date	Created On
05/23/2017	11/13/2018	11/12/2018
Estimated Completion Date	**Next Update Date**	

Revision Description

Revised References section by adding a new service letter.

Revised some terms used in the Description section to be consistent with the new service letter.

Added the delivery date for aircraft line number 7064 in the Status section.

Updated Final Action to state that the UTAS SB and Boeing SL were released.

Updated Milestones to show completion dates for UTAS SB and Boeing SL.

Revised supplier name under Supplier Information section.

This FTD article has been closed.

Applicability

All 737MAX aircraft and 737NG aircraft delivered since April 2013. 737NG aircraft before April 2013 and some 737 Classic aircraft could be affected by installation of a spare unit.

Description

Many operators have reported occurrences where the displays for FLT ALT, LAND ALT, and the outflow valve position indicator (VPI) on the DSP have been identified as blank, too dim, too bright, flashing bright and dim, or having VPI in the wrong position.

Those issues resulted in schedule delays, air turnbacks, and maintenance impacts.

The majority of removed DSPs returned to the supplier for testing resulted in a No Fault Found (NFF) finding.

Background

The P5-6 Cabin Pressurization DSP was redesigned to replace obsolete electronic components. The two LCD displays, lightplate,

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and valve position indicator (VPI) were replaced with an integrated lightplate which provides equivalent functionality using LED technology. New P/N 1019439-1-001 was released in April 2013 with a production cut-in of line number 4413. The same DSP part number is used on the 737MAX.

In April 2015, Hamilton Sundstrand (now UTC Aerospace Systems) and their supplier of the new LED lightplate began investigating this issue and developed ideas for testing. Sensitivity to electromagnetic interference (EMI) was suspected, and this was evaluated by UTC Aerospace Systems (UTAS) lab tests in November 2015. Then in April 2016, Boeing performed on-aircraft testing in order to check for EMI as a possible root cause. The test results by UTAS and Boeing did not clearly explain the display lighting anomalies seen in the fleet.

UTAS performed additional analysis and lab testing to further determine the characteristics of EMI, and to develop ideas for corrective action. In addition to analyzing EMI effects, an inspection and analysis was completed for two electrical relays (R550 and R87), returned from the field from one aircraft, which are in the circuit for power to the DSP. Inspections did not identify a problem with these relays related to the display dimming/blanking issues on that aircraft.

Boeing performed EMI lab testing in Dec 2016, but the data obtained was not sufficient to determine the root cause. A test report was completed in Feb 2017.

Status

In March 2017, Boeing held a Non-Advocate Review (NAR) with UTAS and Korry to review all data associated with this investigation and recommend next steps to determining root cause and corrective action. The NAR group identified root causes within the integrated display faceplate (IDF). The team identified a software issue with the microcontroller clock and a hardware synchronization issue affecting the display drivers. These software and hardware issues result in erratic display behavior that can cause the variety of display anomalies observed in the fleet. A briefing of the RCCA plan was provided during the May 2017 Fleet Team Call, and a project update was provided at the October 2017 FTM.

Root Cause: Improper clock synchronization and control within the Digital Selector Panel lightplate circuitry and software can result in random occurrences of LED and backlighting display anomalies including blanking, flashing, and improper brightness level.

UTAS provided a revised draft Engineering Change Proposal (ECP) on 21-APR-2017. The ECP received Boeing acceptance in mid-July. Boeing PRR (Production Revision Record) 38275-210 was approved August 2017. Production incorporation planned for LN (Line Number) LN7063 (NG) and LN7064 (MAX). LN7063 delivery date was 31-Jul-2018.
LN7064 delivery date was 31-Aug-2018.

Interim Action

Fault Isolation Manual (FIM) 21-31 contains procedures for troubleshooting a faulty DSP. During troubleshooting for the blanking/dimming issue, operators found that the displays may be restored on the ground by resetting pressurization and lighting control circuit breakers. A revision to FIM 21-31 was approved to add a circuit breaker reset procedure to existing DSP troubleshooting procedures. The FIM update was released on 15-Jun-2016. An additional FIM revision was released in Jan 2018 to address the possible condition of display lighting being frozen on full bright.

The Quick Reference Handbook (QRH) Non-normal checklists contain guidance for cabin pressurization system indications such as AUTO FAIL and Cabin Altitude Warning. For the DSP blanking/dimming issue, a non-normal checklist is not included, but general guidance on possible crew action is provided in the QRH Checklist Instructions on page CI.2.2. Due to the in-progress design update, a revision to the QRH is not planned.

In response to operator requests for more information, a Flight Crew Operations Manual Bulletin (FCOMB), on Cabin

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Pressurization Panel Blanking/Dimming Issues associated with P/N 1019439-1-001, was prepared in October 2016 and was released to operators. Bulletins are dated and numbered sequentially for each operator. The FCOMB provides additional guidance to flight crews on recommended actions when a DSP blanking/dimming event occurs. Revision 01 to the FCOMB was released in December 2016.

Final Action

A design revision to the Integrated Display Faceplate (IDF) of the DSP has been completed. The improved DSP part number will be 1019439-2-001.
UTAS released a Service Bulletin (SB 1019439-21-02) to upgrade in-service units when returned for repair.

Boeing issued a Service Letter (737-SL-21-123) to notify operators of the new DSP availability and associated IPC updates.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Boeing FIM Revision: 15-JUN-2016 (completed)

FCOM Bulletin Rev 01: 19-DEC-2016 (completed)

Root Cause Established: 21-APR-2017 (completed)

Fleet Team Update: 9-MAY-2017 (completed)

PRR 38275-210 Committed Date: 30-AUG-2017 (completed)

Production Incorporation (LN7063): 31-JUL-2018 (completed)

Spare Parts Available: August-2018 (completed)

UTAS Service Bulletin Available: 06-SEP-2018 (completed)

Boeing Service Letter Issued: 09-OCT-2018 (completed)

Operator Action

Operators are requested to provide their DSP removal data over the past 12-month period, if not already doing so through the In Service Data Program (ISDP). We also request information be provided when a display anomaly occurs, such as the flight phase, operational conditions during the anomaly and the duration of the anomaly being present.

Part Information

The part number for the DSP is P/N 1019439-1-001 (Boeing Spec No. 10-62231-31). The new part number for the revised DSP will be 1019439-2-001.

Supplier Information

Hamilton Sundstrand

UTC Aerospace Systems (UTAS)

1 Hamilton Road

Windsor Locks, CT 06096-1010

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Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Production Revision Record (PRR)	38275-210	R	05/22/2018
Production Revision Record (PRR)	38958	W	05/06/2013
Flight Crew Operations Manual Bulletin (FCOMB)	TBC-101	01	12/19/2016
Illustrated Parts Catalog (IPC)	21-31		
Fault Isolation Manual (FIM)	21-31		
Fleet Team Digest Article (FTD)	737-FTD-21-12002		
Service Letter (SL)	737-SL-21-074	E	03/31/2015
Fleet Team Economic (FTE)	FTE-21-16009		02/24/2016
Fleet Team Digest Article (FTD)	737NG-FTD-21-16002		
Service Letter (SL)	737-SL-21-123	new	10/09/2018

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Digital Selector Panel	10-62231-31	Hamilton Sundstrand (UTAS)	1019439-2-001
Digital Selector Panel	10-62231-31	Hamilton Sundstrand (UTAS)	1019439-1-001

Related Categories

- CRC - Component Reliability Collaboration : Collaborative process to eliminate component removals in production and the during the first year of new airplane in service
- Fleet Team IdeaXchange : Topic of discussion on the Fleet Team IdeaXchange bulletin board

Attachments:

a) Digital Selector Panel

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